Name : Chitale Ketan Pravin
PRN : 202501100037
Batch : S12

Wordnet dataset

synset_id	word	pos	definition	hypernym	frequency
1	dog	noun	domestic animal	animal	50
2	cat	noun	small feline animal	animal	45
3	car	noun	road vehicle	vehicle	70
4	bus	noun	public transport	vehicle	60
5	tree	noun	woody plant	plant	30
6	flower	noun	flowering plant	plant	25
7	run	verb	move fast	move	80
8	walk	verb	move slowly	move	55

Code : print(df['frequency'].sum())
Output : 415

Code : print(np.median(df['frequency']))
Output : 52.5

Code : print(df.loc[df['frequency'].idxmax()])
Output :

```
synset_id       7
word          run
pos          verb
definition   move fast
hypernym     move
frequency      80
```

word_length   3

Code : print(df.loc[df['frequency'].idxmin()])
Output :synset_id   6
word   flower
pos   noun
definition   flowering plant
hypernym   plant
frequency   25
word_length   6

Code : print(df['pos'].value_counts())
Output :noun   6
verb   2

Code : print(df.groupby('pos')['frequency'].mean())
Output : noun   46.67
verb   67.50

Code :print(df.groupby('hypernym')['frequency'].sum())
Output : animal   95
move   135
plant   55
vehicle   130

Code :
print(df.groupby('hypernym')['frequency'].sum().sort_values(ascending=False))
Output : move   135
vehicle   130
animal   95
plant   55

Code : print(df['hypernym'].value_counts())
Output : animal 2
vehicle 2
plant 2
move 2

Code : print(df['word_length'].mean())
Output : 4.25

Code : print(df.loc[df['word_length'].idxmax()])
Output : word = dog / cat / car / bus / run
length = 3

Code : hist, bins = np.histogram(df['frequency'], bins=4)
print(hist)
print(bins)

Output :[2 2 2 2]
[25. 38.75 52.5 66.25 80.]

Code :print(np.std(df['frequency']))
Output :17.96

Code : print(np.median(df['frequency']))
Output : 52.5

Code : print(df[df['frequency'] > 50])
Output : car, bus, run, walk

Code :print(df.sort_values(by='frequency', ascending=False))
Output : run 80
car 70
bus 60
walk 55

Code :print(df[df['definition'].str.contains("animal")])

Output : dog, cat